

Comparison of BCG, PCG and ECG Signals in Application of Heart Rate Monitoring of the Human Body

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Abstract—The aim of this article is evaluation and comparison of three different signals from three different sensors in application of monitoring the Heart Rate (HR) of the human body. Authors assembled two different types of fiber optic sensors. The first type of sensor uses Fiber Bragg Grating (FBG) and monitoring of heart rate is based on Ballistocardiography (BCG) principle. The second type of sensor utilizes fiber interferometer and monitoring of heart rate is based on Phonocardiography (PCG). Evaluation of signal from the first two types of sensors was acquired by proposed Optical Interrogators. In case of third type (reference signal), heart rate signal was acquired with the use of standard gel electrodes and modules fixed to the monitored person's chest using a real-time monitoring system for an Electrocardiography (ECG) with virtual instrumentation (National Instruments ELVIS II). The experimental tests were carried out on six subjects of both sexes in a laboratory conditions with their written consent. The obtained data were compared objectively by the Bland-Altman method.

Keywords—biomedical signal processing; fiber-optic sensors; fiber Bragg grating; heart rate; interferometer;

I. INTRODUCTION

Recent years we can see an increasing in the utilization of fiber-optic sensors and instruments for a variety of emerging biomedical applications [1]–[7]. The characteristic features of fiber-optic sensors are their immunity to electromagnetic interference (EMI) and independence from an active power supply. Also, these sensors can be used with other electronic equipment (because they don't generate any electric noise that may compromise the quality of biomedical signals and potentially lead to patient safety concerns). Thanks to these attributes (and also due to their flexibility, improved functionality, and reliability), fiber-optic sensors is gaining more popularity in biomedical applications. The authors have developed a two fiber-optic measurement probes encapsulated in polydimethylsiloxane [8]–[9] that allows for monitoring the Ballistocardiography (BCG) and Phonocardiography (PCG) signals evoked by human body. The BCG is a measure of ballistic forces on the heart. The BCG curve is a graphical representation of repetitive motions of the human body arising from the sudden ejection of blood into the great vessels

with each heart beat [10]. PCG describes its mechanical (acoustic) activity. This classification is associated with a different technique of its measurement. PCG is based on recordings of acoustic signals by means of a microphone. PCG signal is composed of two main acoustic phenomena the first (S_1) and the second (S_2) heart sounds. The S_1 (systolic) is associated with the closure of bicuspid and tricuspid valve at the beginning of systole and S_2 (diastolic) is induced by the closure of semilunar valves [11]. BCG, as well as PCG has obvious advantages such as non-invasiveness, electrically contactless, as well as simple design of measurement systems. ECG is a method that measures the electrical activity of the heartbeat. With each beat, an electrical impulse travels through the heart. This impulse causes the muscle to squeeze and pump blood from the heart. A normal heartbeat on ECG will show the timing of the top and lower chambers [12].

The study of comparing the heart rate signals from various types of sensors are not too many published, for example article [13] published in 1969 brings introductory information from synchronous recordings of BCG, ECG, Carotid Pulse Wave, and PCG in acute pharmacological trials, but BCG and PCG signals were not obtained by means of fiber-optic sensors. In article [14] the Heart Rate Variability (HRV) indices of simultaneously acquired ECG and BCG signals are compared and found that the BCG signal can provide satisfactory results as that of ECG, so for long term monitoring BCG can be used instead of ECG. In article [15] a new approach used to detect specific parameter and relations between three biomedical signals that used in clinical diagnosis. These include the phonocardiography (PCG), electrocardiography (ECG) and Photoplethysmography (PPG) - the carotid pulse related to the position of electrode.

The aim of this article is more detailed evaluation and comparison signals from fiber-optic sensors against reference ECG sensor in the application of monitoring the Heart Rate (HR) of the human body. The experimental tests were carried out on six subjects of both sexes (three men and three women) in a laboratory conditions and obtained data were compared

objectively by the Bland-Altman method [16].

II. MEASUREMENT METHODS

The first type of sensor uses Fiber Bragg Grating (FBG) and monitoring of heart rate is based on Ballistocardiography (BCG) principle. The FBG is characterized by a periodic change of the refractive index in the fiber's core (n_1 , n_3), see Fig. 1. Symbol n_2 represents the refractive index of the fiber cladding. Single-point FBGs can be connected together in a successive order, thereby obtaining a multi-point sensor within one optical fiber. The easiest method for the resolution of individual sensors is the use of wavelength-division multiplex [17]–[18]. Used type of FBG is uniform FBG with polyimide protection with the Bragg wavelength of 1554.1104 nm, the width of the reflecting spectrum of 2.3345 nm, and a reflectivity of 95.5 %.

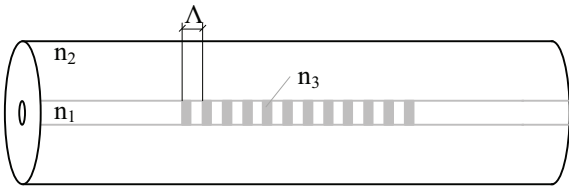


Fig. 1. Fiber Bragg Grating structure.

If is project a beam of light from a Light Emitting Diode (LED) into an optical fiber with FBG, a narrow spectral part of the beam is reflected. This reflected wavelength is called the Bragg wavelength λ_B . The size of λ_B is given by the equation:

$$\lambda_B = 2n_{eff}\Lambda, \quad (1)$$

where λ_B represents Bragg wavelength, n_{eff} represents the effective refractive index of the optical fiber and Λ represents the period of refractive index changes in the optical fiber.

Deformation dependence is given the central λ_B and parameter values. To determine individual sensitivities, normalized temperature and deformation coefficients used. Normalized temperature coefficient is given by

$$\frac{1}{\lambda_B} \frac{\Delta\lambda_B}{\Delta T} = 6.678 \times 10^{-6} \quad ^\circ C^{-1} \quad (2)$$

and normalized deformation coefficient is given by [19]

$$\frac{1}{\lambda_B} \frac{\Delta\lambda_B}{\Delta\epsilon} = 0.78 \times 10^{-6} \quad \mu\epsilon^{-1} \quad (3)$$

where λ_B represents Bragg wavelength, $\Delta\lambda_B$ represents change of Bragg wavelength and ΔT resp. $\Delta\epsilon$ represents temperature resp. deformation change.

The second type of sensor uses dual coupler fiber-optic Mach-Zehnder interferometer (power ratio 1:1) and monitoring of heart rate is based on Phonocardiography (PCG). Interferometry is an optical method that can monitor the phase difference between two optical beams which pass through

similar optical paths. Phase shift arises in the interferometer and is given by:

$$\phi = 2\pi L \frac{n}{\lambda} = kLn, \quad (4)$$

where n represents the refractive index of the core, L represents the length of optical fiber, k represents the size of the wave vector and λ represents the wavelength of the radiation source. The output intensity of the interferometer is given by:

$$I = \frac{I_0\alpha}{2}(1 + \cos \Delta\phi). \quad (5)$$

In this relation α represents the optical loss of the interferometer, I_0 represents light intensity on the input of coupler and $\Delta\phi = (\phi_s - \phi_r)$ represents the phase difference between the measuring and reference arms of the fiber-optic interferometer. Intensity on the output of photodetector creates electrical current [20] which is given by:

$$i = \varepsilon \cdot I_0 \cdot \alpha \cdot \cos(\phi_d + \phi_s \sin \omega t), \quad (6)$$

where ε is the sensitivity of the photodetector (photodetector responsivity), ϕ_d is slowly varying phase shift, ϕ_s is duration of the amplitudes and ω is the frequency.

A. Proposed Schematic diagram of Grating Optical Interrogator (GOI)

Resulting signal from FBG sensor is further processed by a spectral evaluation of measured signal by proposed Grating Optical Interrogator (GOI), see Fig. 2. Reflected signal from the measurement channel is projected through the three-port optical circulator into an optical spectrum analyzer (OSA). We used an optical spectrum analyzer with sample rate 200 Hz. The acquired signal is then pre-processed in the Digital Signal Processing (DSP) Unit. The Electronic Control Unit (ECU) controls parameters such as output power of used LED (Light-Emitting Diode) or a sampling rate of the OSA. The digitalized signal with the required information about heart rate is shown on PC using the application created in LabView (2015, National Instruments, Austin, Texas, USA).

B. Proposed Schematic diagram of Interferometric Optical Interrogator (IOI)

Resulting electric signal from interferometric sensor is further processed by proposed Interferometric Optical Interrogator (IOI). Schematic diagram of Interferometric Optical Interrogator (IOI) and electrical signal processing is shown on Fig. 3. A tunable laser source with the output power of 1 mW was used as a source of radiation. An optical isolator is an optical component which allows the transmission of light in only one direction. It is used to prevent unwanted feedback into an optical oscillator, in our case a laser cavity. The Lead Selenide (PbSe) photodetector (PD) registers the signal which arises as a result of interference of optical beams from the reference and measuring arm of the interferometric sensor, and it changes signal into a measurable electric current. The signal processing devices (Signal processing) uses a high-pass filter (HP) with a cutoff frequency at 8 Hz via 50 Ω coaxial

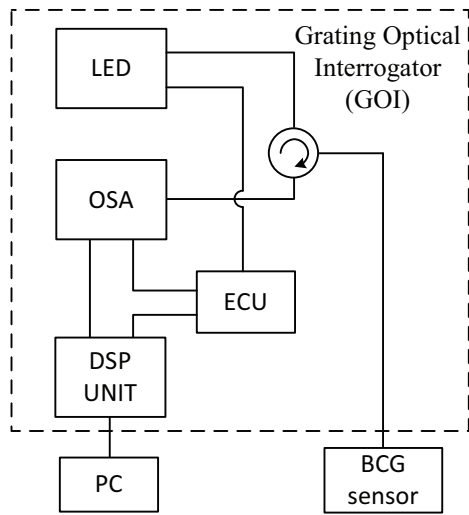


Fig. 2. Schematic diagram of Grating Optical Interrogator (GOI).

cable to suppress DC component [21], amplifier, and analog-digital converter in the form of measuring card labeled NI USB 6210 (National Instruments). The sampling rate is 50 kS/s. The digitalized signal with the required information about heart rate is shown on PC using the application created in LabView (2015, National Instruments, Austin, Texas, USA).

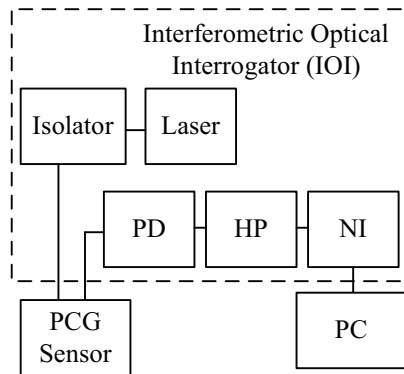


Fig. 3. Schematic diagram of Interferometric Optical Interrogator (IOI)

III. RESULTS

A verification study was carried out on a group of six subjects (three men (M1-M3) and three women (F1-F3) in laboratory conditions. Each of the experiments included simultaneous recording of the FBG sensor (BCG signal), interferometer sensor (PCG signal) and ECG sensor. Based on the detection of peaks and the calculation of time intervals between these peaks the heart rate was determined. Verification included over 4500 heartbeats. The total time was 60 min 49 s. The reference heart signal was acquired with the use of standard gel electrodes and modules fixed to the monitored person's chest using a real-time monitoring system for an Elec-

trocardiography (ECG) with virtual instrumentation (National Instruments, ELVIS II, 2015).

Both types of presented fiber-optic sensor for measuring the heart rate of the human body were encapsulated inside a polymer polydimethylsiloxane (PDMS) in the consistent laboratory conditions (vacuum box, 24 °C, the same weight of PDMS).

Firstly, the acquired data of respiratory rate (RR) in case of FBG sensor were processed by using a Butterworth second-order band-pass filter with corner frequencies $FL = 0.75$ Hz and $FH = 5$ Hz, respectively. The respiration rate (below 0.75 Hz - it corresponds to a speed of 45 breaths) and the unwanted noise signal (>5 Hz) were filtered out by the Butterworth second-order low pass filter. To compensate the drift (the fluctuation of zero isoline, breathing), we selected again a lower marginal frequency of 0.5 Hz. An upper marginal frequency is defined especially by the phonocardiography frequency of heart to the value 400 Hz [22]–[23].

Figure 4 shows example of the 60-second (a) RR trace and (b) HR trace from the BCG signal acquired by FBG sensor of test subject M3.

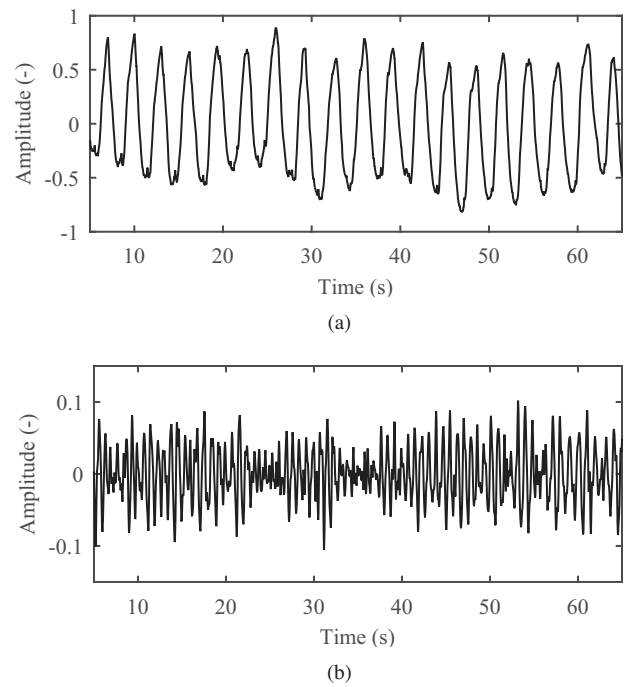


Fig. 4. 60-second trace of (a) RR and (b) HR from the BCG signal acquired by FBG sensor of test subject M3.

Figure 5 shows example of the 60-second (a) RR trace and (b) HR trace from the PCG signal acquired by interferometric sensor of test subject M3.

The key experimental results of heart rate measurement from each used sensors are summarized in Table 1. To compare the differences between the reference ECG signal and signal from the FBG sensor and interferometric sensor, the Bland-Altman method was utilized. Bland and Altman

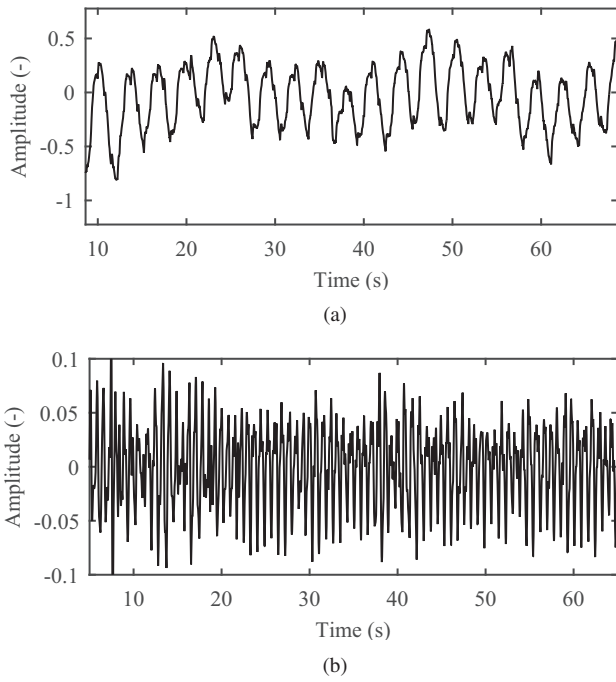


Fig. 5. 60-second trace of (a) RR and (b) HR from the PCG signal acquired by interferometric sensor of test subject M3.

analysis respective plot allow us to the identification of any systematic difference between the measurements (that means measured value and reference value). The mean difference is the estimated bias and the SD (Standard deviation) range of the differences measures the random fluctuations around this mean. So it is common to determine 95 % limits of agreement for our comparison (average difference ± 1.96 SD of the difference).

TABLE I. THE KEY EXPERIMENTAL RESULTS OF HEART RATE MEASUREMENT.

Subject	Rec. time (s)	Nos BCG sensor	Samples in ± 1.96 SD (%)	Nos PCG sensor	Samples in ± 1.96 SD (%)
M1	720	824	95.02	831	95.39
M2	610	754	94.56	758	94.69
M3	560	668	95.66	664	95.46
F1	550	790	94.12	788	94.05
F2	590	826	96.13	817	95.84
F3	620	844	94.91	849	95.02

For the entire data set from PCG sensor, 95.07 % of the values lie within the 1.96 SD range for the HR determination, see Fig. 6a. For the entire data set from FBG sensor, 95.06 % of the values lie within the 1.96 SD range for the HR determination, see Fig. 6b. If we compare results acquired by the Bland-Altman method, we can statement, that interferometric sensor in the application of heart rate monitoring of the human body is more accurate (0.01 %) against FBG sensor.

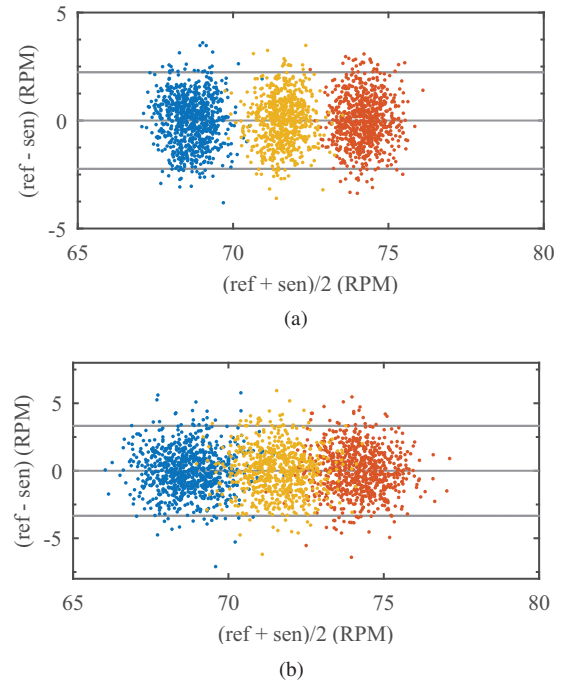


Fig. 6. Statistical analysis of heart rate using the Bland-Altman method for (a) interferometric (PCG) sensor and reference ECG sensor (a) FBG sensor and reference ECG sensor.

IV. CONCLUSION

The conclusion goes here. In this article, we have reported methods for detecting the HR trace of human body from the BCG and PCG signal acquired by the MRI-compatible fiber Bragg grating and fiber interferometric sensors. The experimental tests were carried out on six subjects of both sexes in a laboratory conditions and obtained data were compared objectively by the Bland-Altman method against reference ECG signal. For the entire data set from FBG sensor, 95.06 % of the values lie within the ± 1.96 SD range, and for the entire data set from interferometric sensor, 95.07 % of the values lie within the ± 1.96 SD range for the HR determinations. Based on the Bland-Altman analysis the interferometric sensor in the application of heart rate monitoring of the human body is more accurate (0.01 %) against FBG sensor.

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